SRINIVAS UNIVERSITY
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MODULE 5

NATURAL LANGUAGE PROCESSING

INDIVIDUAL TASK

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INDIVIDUAL TASK

Chatbot Sales Conversation: Interact with a customer-service bot. Analyze how well it interprets intent and closes a "deal."

Chatbot Sales Conversation – NLP Based Customer Interaction

1. Introduction

Natural Language Processing (NLP) is a branch of Artificial Intelligence that enables machines to understand, interpret, and respond to human language. Chatbots are one of the most common applications of NLP used in customer service and sales. These bots interact with users, identify their intent, and provide appropriate responses.

In this task, a chatbot sales conversation is analyzed to understand how NLP interprets user input and helps close a sales deal. The chatbot identifies keywords, detects intent, and generates suitable responses.

2. Objective

The objective of this task is:

- To interact with a chatbot
- To analyze how NLP understands user queries
- To evaluate chatbot response accuracy
- To observe intent recognition
- To analyze whether chatbot can close a sale

3. Tools Used

- AI chatbot (ChatGPT / Dialogflow / Website bot)
- NLP-based conversation system
- Internet browser

4. Chatbot Sales Conversation Example

User: I want to buy a laptop
Chatbot: Sure! What is your budget?

User: Around 50000
Chatbot: I found laptops under 50000. Do you prefer HP, Dell, or Lenovo?

User: Dell
Chatbot: Dell Inspiron 15 is available for ₹48,000. Would you like to order?

User: Yes
Chatbot: Your order has been placed successfully.

5. NLP Working in Chatbot

The chatbot uses NLP in the following steps:

Step 1: Input Processing

User types a message.
Example: "I want to buy laptop"

Step 2: Tokenization

Sentence split into words
I | want | buy | laptop

Step 3: Intent Recognition

Intent detected: Purchase product

Step 4: Entity Extraction

Entity detected:
Product = Laptop

Step 5: Response Generation

Chatbot asks for budget

Step 6: Context Understanding

Chatbot remembers previous message

Step 7: Final Action

Chatbot completes purchase

6. Advantages of Chatbot Sales Using NLP

- 24/7 customer support
- Fast response time
- Automatic product recommendation
- Reduces human workload
- Improves customer satisfaction

- Increases sales conversion

7. Limitations

- Cannot understand complex sentences
- May give wrong suggestions
- Limited contextual memory
- Depends on training data

8. Result

The chatbot successfully interpreted user intent and completed the sales process. NLP helped in understanding user requirements and generating relevant responses.

9. Conclusion

This task demonstrated how NLP is used in chatbot sales conversations. The chatbot identified user intent, extracted entities, and generated appropriate responses. The chatbot successfully completed the sales transaction. Hence, NLP plays an important role in automated customer interaction and sales.